

Integral transforms and applications LTP- 3-0-0, CRD- 3

Prerequisites: Ordinary and Partial Differential Equations, Complex analysis

Introduction to integral transform; Why to learn Integral Transform? Definition of Integral Transform; From Fourier series to Fourier integral (Fourier integral representation); Fourier sine and cosine integral; Fourier sine and cosine transform; The exponential Fourier integral and Fourier transform. Properties of Fourier transforms (linearity, shifting, scaling and others); Convolution theorem and Parseval Relation; Fourier Transforms in Two or More Dimensions; Application of Fourier Transforms in Solving Ordinary and Partial Differential Equations. Definition of Laplace Transform; Derivation of Laplace Transform from Fourier Integral Theorem, Linearity property; Conditions for existence of Laplace Transform. Laplace Transform of some elementary functions (by definition). Different properties of Laplace transform; Laplace inversion (a) by using properties (b) by using contour integration. Initial and Final value theorems and proofs. Laplace Transform of some special functions. Convolution theorem; Evaluation of integrals by Laplace Transform; Solution of integral equation by Laplace Transform. Application of Laplace Transforms in Solving Ordinary and Partial Differential Equations with initial and boundary conditions.

Definition of Hankel Transform; Properties of Hankel Transform; Hankel Transform of some simple functions; Applications

Definition of Mellin Transform; Theorem on Inversion of Mellin Transform; Properties of Mellin Transform; Mellin Transform of Some elementary Functions

Introduction to Z transform; Properties of Z transform; Inverse Z transform

Books:

1. L. Debnath and D.D. Bhatta, Integral Transforms and Their Applications, Book World Enterprises, 2006.
2. M. Ya. Antimirov, A.A. Kolyshkin, R. Valliancourt, Applied Integral Transforms, CRM Monograph Series, American Mathematical Society, 2007.
3. A.D Poularikas, The Transforms and Applications Handbook, CRC Press, 1996.
4. S. Banerjea and B. N. Mandal, Integral Equations and Integral Transforms, Springer, 2023.